Dr. Anika Sharma

Scientist E — Atmospheric Instrumentation & Applied AI

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SKILLS

Avionics

- MIL-STD-1553 / ARINC 429
- Avionics integration

Scientific instrumentation

- Environmental qualification testing
- Sensor calibration
- Radiosonde and upper-air systems

Embedded systems

- Embedded C
- ESP32 / STM32 microcontrollers

Computer vision

- Image super-resolution
- Object detection

Drones & UAV systems

- UAV payload integration

Edge AI

- NVIDIA Jetson deployment

AI / machine learning

- Deep learning
- PyTorch

Procurement & administration

- Technical procurement (GeM, GFR)

IoT

- MQTT and Node-RED
- InfluxDB and Grafana

Model optimisation

- TensorRT and ONNX
- Pruning and knowledge distillation

Software & tools

- Python
- Docker and Linux

Technical leadership

- Team leadership

GPU computing

PROFESSIONAL SUMMARY

Electronics and avionics engineer with twelve years of airborne-systems integration and qualification testing, followed by ten years leading scientific instrumentation and edge-AI development in government service. Experienced in MIL-STD-1553 test rigs, UAV payload electronics, sensor telemetry, embedded GPU deployment and technical procurement.

AREAS OF EXPERTISE

R&D in avionics systems · 12 years

Twelve years of mission-computer and sensor integration, interface testing (MIL-STD-1553, ARINC 429) and environmental qualification for airborne platforms at Halcyon Avionics Design Bureau.

Edge AI for aerial imagery · 5 years

Research on super-resolution and model compression (pruning, knowledge distillation) for real-time deployment on NVIDIA Jetson devices; publications and a deployed telemetry prototype.

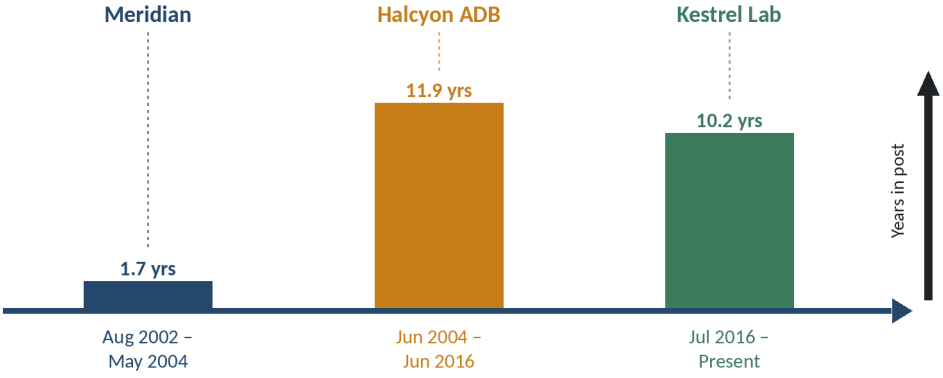
Upper-air instrumentation · 10 years

Planning, testing and maintenance of radiosonde ground systems and upper-air instruments; division workshop management; technical evaluation of procurements.

IoT monitoring systems · 4 years

Design of MQTT, Node-RED, InfluxDB and Grafana pipelines for monitoring ground-station equipment.

CAREER TIMELINE



- Meridian Electronics Limited — Graduate Engineer (Aug 2002 – May 2004): Automated test equipment for avionics units
- Halcyon Avionics Design Bureau — Senior Design Engineer (Jun 2004 – Jun 2016): Mission-computer integration and environmental qualification
- Kestrel Atmospheric Instrumentation Laboratory — Scientist E (Jul 2016 – Present): Upper-air instruments, workshop, IoT and edge-AI programmes

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PROFESSIONAL EXPERIENCE

Scientist E — Jul 2016 – Present

Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory, New Delhi

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CUDA / GPU computing

Bars show my own assessment of proficiency (five steps; five = expert).

EDUCATION

Ph.D., Electrical and Electronics Engineering
Northfield Institute of Science and Technology
2021 – ongoing
Thesis: Lightweight deep networks for real-time enhancement of aerial imagery

M.Tech., Avionics
Coastal Institute of Aeronautical Engineering
2010
Grade: CGPA 8.6/10
Thesis: Fault-tolerant data acquisition for airborne test instrumentation

B.E., Electronics and Communication Engineering
Northfield Institute of Science and Technology, Northfield University
2004
Grade: First Class with Distinction

Responsible for upper-air observation instruments, the division workshop and the laboratory's IoT and AI/ML activities.

Tools: Python, PyTorch, MQTT, Node-RED, InfluxDB, Grafana, Docker

- Lead a four-member AI/ML group developing lightweight super-resolution and detection models for UAV imagery, deployed on NVIDIA Jetson devices with TensorRT.
- Led acceptance testing and commissioning of radiosonde ground-receiving systems at three observing stations, writing the test protocol and the defect-tracking procedure.
- Designed and deployed an IoT monitoring network for laboratory infrastructure (ESP32 sensor nodes, MQTT, Node-RED, InfluxDB and Grafana dashboards) covering temperature, humidity and power quality.
- Prepared technical specifications and served on bid-evaluation committees for instrument procurement through the Government e-Marketplace in line with the General Financial Rules.
- Authored standard operating procedures for pre-flight checks and calibration of upper-air instruments used across the division.

Additional charge: **Workshop In-charge**, Upper Air Division Workshop — Apr 2019 – Present

Additional charge of the division workshop supporting instrument repair, fabrication and calibration.

- Supervise technicians and plan preventive maintenance for workshop test equipment.
- Introduced a job-card and spares-inventory system for instrument repairs and calibration work.

Nodal officer: **Nodal Officer, IoT and AI/ML Initiatives** — Jan 2021 – Present

Senior Design Engineer — Jun 2004 – Jun 2016

Avionics Integration Group, Halcyon Avionics Design Bureau, Bengaluru

Integration and qualification testing of airborne electronics and UAV payloads.

Tools: MIL-STD-1553, ARINC 429, LabVIEW, C, MATLAB

- Developed payload interface electronics for a fixed-wing UAV electro-optical sensor and supported its ground and flight trials.
- Integrated mission-computer and display units on MIL-STD-1553 avionics buses and built the automated test rig used for bench validation.
- Planned environmental qualification (MIL-STD-810 vibration, temperature and humidity) for line-replaceable units and documented the test results.
- Mentored graduate engineer trainees on avionics bus protocols and test documentation.

Graduate Engineer — Aug 2002 – May 2004

Test Systems Group, Meridian Electronics Limited, Bengaluru

Built automated test equipment for avionics line-replaceable units.

Tools: LabVIEW, C

- Developed LabVIEW test sequences for power-supply and interface boards of avionics units.

PROJECT PORTFOLIO

Edge super-resolution pipeline for UAV imagery *(Ongoing)* — Sep 2021 – Present

Principal investigator, Kestrel Atmospheric Instrumentation Laboratory

Research prototype that enhances low-resolution UAV video in real time on embedded GPUs using pruned and distilled deep networks.

- Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment.
- Converted models through ONNX to INT8 TensorRT engines, achieving 3.1× faster inference on Jetson Xavier NX.

Technologies: NVIDIA Jetson Xavier NX, NVIDIA Jetson Orin Nano, PyTorch, TensorRT, ONNX, Docker, Structured pruning, Knowledge distillation, INT8 quantisation

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Radiosonde ground-station acceptance test protocol — Jan 2018 – Mar 2019

Coordinator

Challenge: Station acceptance testing varied between sites.

Outcome: A common acceptance protocol and checklist adopted for new ground-receiving systems.

IoT monitoring of laboratory infrastructure — Feb 2020 – Jun 2022

Project lead

Challenge: Instrument rooms lacked continuous environmental and power-quality monitoring.

Approach: Deployed ESP32 sensor nodes publishing over MQTT to a Node-RED flow, InfluxDB time-series storage and Grafana dashboards with threshold alerts.

Outcome: Continuous monitoring with automated alerts replaced manual twice-daily readings.

Technologies: ESP32, Environmental sensors, MQTT, Node-RED, InfluxDB, Grafana

PUBLICATIONS

[1] **A. Sharma** and R. Iyer, "Knowledge-distilled lightweight networks for drone image super-resolution," accepted for publication in Example Transactions on Aerial Imaging.

[2] **A. Sharma**, K. Rao and R. Iyer, "Benchmarking embedded GPUs for real-time image enhancement," Example Symposium on Applied AI, pp. 88-93, Jan 2025.

[3] **A. Sharma** and R. Iyer, "Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs," Journal of Embedded Vision Systems, vol. 12, no. 3, pp. 45-58, Mar 2024. doi: 10.5555/jevs.2024.0123

[4] **A. Sharma**, P. Rao and S. Gill, "An IoT telemetry framework for upper-air ground stations," Proceedings of the Example Conference on Instrumentation (ECI 2023), pp. 112-117, Sep 2023.

[5] **A. Sharma** and S. Verma, "Edge telemetry dashboards for upper-air ground stations," Proceedings of the Example Conference on Instrumentation, Paper 17, Feb 2022.

[6] **A. Sharma**, P. Menon and R. Iyer, "A low-cost GNSS receiver chain for radiosonde tracking," Example Journal of Atmospheric Instrumentation, vol. 8, no. 2, pp. 101-112, Jun 2021. doi: 10.5555/ejai.2021.0042

AWARDS & HONOURS

Best Paper Award — 2023

Example Conference on Instrumentation (ECI 2023)

REFEREES

Dr. R. Iyer (fictional)

Professor of Electronics, Example Institute of Technology

Research collaborator and co-author

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